

STANDARD OPERATING PROCEDURE

SAP Data Export and Upload to Google Sheets Automation System

Document Control

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1 PURPOSE

This Standard Operating Procedure (SOP) defines the end-to-end process for:

- Exporting the correct data from SAP with all required columns included.
- Preparing and validating the exported file before upload.
- Uploading the file into the Google Sheets Automation System correctly for Monthly Production, Monthly Sales Quantity, and Sale Price sections.

The objective of this SOP is to ensure that every user, regardless of department or experience level, follows a consistent, error-free process that produces accurate data in the target Google Sheet every month.

2 SCOPE

This SOP applies to:

- All staff members responsible for exporting SAP data and uploading it to the Google Sheets Automation System.
- The following three data sections: **Monthly Production**, **Monthly Sales Quantity**, and **Sale Price**.
- The following departments: **Operations**, **Finance**, **Logistics**, and **HR**.
- The Indian Financial Year cycle: **April to March (YYYY–YY format, e.g., 2025–26)**.

This SOP does **not** cover:

- Direct editing of the target Google Sheet.
- SAP system administration or configuration.
- Backend code changes to the automation system.

3 DEFINITIONS AND ABBREVIATIONS

Table 1: Definitions and Abbreviations

Term / Abbreviation	Definition
SAP	Systems, Applications and Products - the ERP software used by the organization
MB51	SAP transaction code for Material Document List (used for Production data)
FAGLL03H	SAP transaction code for List of Billing Documents (used for Sales data)
FAGLL03H	SAP transaction code for List of Billing Documents (used for Sale Price)
GL Account	General Ledger Account - financial classification code used for sales/revenue entries

Term / Abbreviation	Definition
Material Code	Unique numeric identifier assigned to each product/material in SAP
Movement Type	3-digit SAP code for type of stock movement
Plant Code	4-digit code identifying manufacturing or storage location
UoM	Unit of Measure (e.g., MT = Metric Ton)
MT	Metric Ton - standard unit of measure for all quantity-based sections
FY	Financial Year (Indian format: April to March)
Web App	Google Apps Script-based upload interface for this system
Target Sheet	Google Sheet where processed data is written
Operational Data Tab	Specific sheet tab in the Target Sheet where all data is stored
Unknown Code	GL Account or Material Code in the uploaded file not registered in system configuration rules

4 ROLES AND RESPONSIBILITIES

Table 2: Roles and Responsibilities

Role	Responsibility
Data Entry Operator	Export SAP file with all required columns, validate the file locally, and upload via the Web App.
Section Supervisor	Verify that uploaded values in the Google Sheet match expected monthly figures before sign-off.
Finance / Operations Lead	Review final Google Sheet data at month-end and flag any discrepancies.
System Administrator	Maintain Web App deployment, update configuration rules when SAP master data changes, and manage user access.
Senior Software Developer	Handle escalations requiring code-level fixes and add new categories or sections when required.

5 SYSTEM REQUIREMENTS

Before starting any export or upload, confirm the following:

Table 3: System Requirements

Requirement	Details
SAP Access	Active SAP login with authorization for MB51 or FAGLL03H as applicable.
Browser	Google Chrome (recommended), Microsoft Edge, Mozilla Firefox, or Safari - latest version.
Internet Connection	Stable connection required throughout upload.
Google Account	Google account with edit access to the Target Sheet.
Web App URL	Production Web App URL provided by System Administrator.
File Format	Exported SAP files must be .xlsx, .xls, or .csv only.
File Size	Maximum recommended file size: 5 MB per file; larger files must be split.
Target Google Sheet	Must exist, have an “Operational Data” tab, and be shared with edit access for the uploading user.

6 SAP EXPORT REQUIREMENTS - WHAT TO INCLUDE

The upload system reads specific column headers from the exported file to identify and process the correct data. If any required column is missing, the upload will either fail or produce incorrect results.

6.1 Required Columns for Monthly Production (MB51 Export)

Table 4: Required Columns: Monthly Production

#	Required Column Header (Exact or Partial Match)	Purpose
1	Quantity or Qty	Production quantity in MT that will be summed.
2	Unit of Measure or Unit or UOM	Used to filter only MT rows and exclude other units.
3	Movement Type or Movement or MVT	Used to include only allowed movement types (e.g., 101, 102).
4	Plant	Used to filter by correct manufacturing plant.
5	Material or Material Code or MATNR	Used to identify and validate materials against configured ranges.

Optional but recommended columns: Posting Date, Document Number, Storage Location, Batch Number.

6.2 Required Columns for Monthly Sales Quantity (FAGLL03H / MB51 Export)

Table 5: Required Columns: Monthly Sales Quantity

#	Required Column Header	Purpose
1	Quantity or Billed Quantity	Sales quantity in MT to be summed.
2	Unit of Measure or Unit or UOM	Used to include only MT rows.
3	G/L Account or GL Account or G/L	Used to match rows to correct sales category.
4	Material or Material Code or MATNR	Required if material exclusion rules apply.

6.3 Required Columns for Sale Price (FAGLL03H Export)

Table 6: Required Columns: Sale Price

#	Required Column Header	Purpose
1	Quantity or Billed Quantity	Quantity in MT used as denominator for price calculation.
2	Unit of Measure / Unit / UOM	Used to include only MT rows for quantity.
3	G/L Account / GL Account / G/L	Used to match rows to correct price category.
4	Net Value / Net Amount / Currency Value / Company Code Currency Value / Amount / Value	Revenue amount used as numerator for price calculation.
5	Material / Material Code / MATNR	Required for material exclusion rules.

Sale Price is computed as:

$$\text{Sale Price} = \frac{\text{Total Net Value (MT rows)}}{\text{Total Quantity (MT rows)}}$$

6.4 Critical Rules for All Exports

Table 7: Critical Export Rules

Rule	Requirement
Unit of Measure	All rows intended for upload must have MT in UoM. Rows with other units are ignored. TON, TO, and T are mapped to MT.
No Subtotal / Total Rows	Do not include SAP subtotal or grand total rows. The system tries to skip rows with "TOTAL", "RESULT", or "*", but exporting without them is safer.
No Page Headers or Footers	Remove SAP report title rows and page-break headers from the middle of exports.

Rule	Requirement
GL Accounts - No Leading Zeros	Leading zeros are stripped automatically; configuration must store GLs without leading zeros.
Material Codes - Numeric	Material codes are treated as numeric with leading zeros removed.
Single Sheet Tab Only	Data must be on the first sheet tab only; only this tab is read.
Date Range Must Match	Export data for exactly one calendar month.

7 STEP-BY-STEP SAP EXPORT INSTRUCTIONS

7.1 Monthly Production Export (MB51)

Steps 1–7 describe logging into SAP, running MB51 with correct filters, adjusting layout to include required columns, and exporting to Excel (.xlsx) or .csv, then verifying headers and data as per Section 6.

7.2 Monthly Sales Quantity Export (FAGLL03H or MB51)

Steps 1–7 describe using FAGLL03H, selecting billing date range and sales organization, ensuring layout includes GL Account, Material, Billed Quantity, UoM, Net Value, exporting to spreadsheet, and verifying required columns.

7.3 Sale Price Export (FAGLL03H)

The Sale Price upload typically uses the same file as Monthly Sales, provided it has both quantity and net value columns; otherwise, a supplementary FAGLL03H or G/L report export is required with GL, quantity, UoM, and net value.

8 FILE PREPARATION BEFORE UPLOAD

8.1 File Preparation Checklist

Table 8: File Preparation Checklist

#	Check	What to Verify	Action if Failed
1	File Format	File saved as .xlsx, .xls, or .csv	Re-export from SAP in supported format.
2	File Size	File under 5 MB	Split file by plant/date and upload in parts.
3	Single Sheet Tab	Data on first sheet tab only	Copy data to Sheet1 and delete others.
4	Header Row Present	Column headers in first non-blank row	Delete title rows above header.
5	No Merged Cells	No merged cells in header/data rows	Unmerge all cells.

#	Check	What to Verify	Action if Failed
6	No Mid-Sheet Page Breaks	No repeated headers or page titles	Remove all repeated header rows.
7	No Subtotal or Total Rows	No rows like “Total”, “Result”, or starting with *	Delete subtotal and total rows.
8	Quantity Column Numeric	Quantity column numeric	Convert text to numbers.
9	UoM Column Present	Column header contains “Unit” or “UOM”	Re-export with this column.
10	GL Account Present (Sales/Price)	GL Account column exists	Re-export with GL column.
11	Material Column Present	Material / MATNR column exists	Re-export with Material column.
12	Movement Type Present (Production)	Movement Type / Movement column exists	Re-export with Movement Type.
13	Plant Column Present (Production)	Plant column exists	Re-export with Plant column.
14	Correct Month’s Data	Dates within target month	Re-export with correct date range.
15	No Password Protection	File not password-protected	Remove password and re-save.

8.2 Validating Unit of Measure

Check UoM column and ensure values for relevant rows are MT (or TON/TO/T). Rows with KG, PC, NOS, EA, or others are automatically excluded; do not delete them unless they belong to a different dataset entirely.

9 STEP-BY-STEP UPLOAD PROCEDURE

9.1 Accessing the Web Application

Steps 1–4: open browser, go to Web App URL, sign in with organizational Google account that has sheet access, and view form fields: Google Sheet URL, Sheet Tab, Section, Category checkboxes, Financial Year, Month, and File Upload Area.

9.2 Connecting the Target Google Sheet

If first upload in current FY, paste Target Sheet URL, click Validate, wait for “Connected” message and populated tab list. If URL already saved, it auto-loads; use “Change Sheet URL” if a different sheet is needed.

9.3 Selecting Upload Parameters

Choose the correct Section, then select relevant Category checkboxes; select FY and Month carefully since uploads to a wrong month are not auto-reversible.

9.4 Adding the File

Add the prepared SAP export via drag-and-drop or file picker; verify file name, size (≤ 5 MB), and format (.xlsx/.xls/.csv); remove and re-add if incorrect.

9.5 Submitting the Upload

Click Upload, observe progress indicator, and wait. Approximate processing times range from 3–5 seconds (≤ 500 KB) up to 20–40 seconds (3–5 MB). On completion, a Success, Warning, or Error result appears with required next actions.

10 EXCEPTION HANDLING

This section lists all known exceptions and corrective actions.

10.1 File-Related Exceptions

Exceptions F-01 to F-06 cover unsupported format, empty file, header not found, file too large (timeout), password-protected file, and wrong month data, each with error message, cause, corrective action, and escalation guidance.

10.2 Data Validation Exceptions

Exceptions D-01 to D-08 cover unknown GL accounts, unknown material codes, zero written value, higher/lower than expected values, unrealistic Sale Price, and data written to wrong month or category.

10.3 Upload and Mapping Exceptions

Exceptions U-01 to U-06 cover file-type mismatches between section and export, missing category/month in sheet, backend mapping errors, and duplicate uploads overwriting existing cells.

10.4 System and Access Exceptions

System-level issues include invalid sheet URL or permission denied, Web App not loading, authorization prompts, permission errors during upload, and Drive API quota exceeded, with clear user and admin actions.

11 DO'S AND DON'TS

11.1 DO's

Table 9: DO's

#	Do
1	Always export SAP data for exactly one month at a time.
2	Always include all required columns listed in Section 6.
3	Always validate your Google Sheet URL at the start of every new FY.

#	Do
4	Always verify the written value in the Target Sheet immediately after each upload.
5	Always use the naming convention in Section 8.
6	Always remove page-break headers, subtotal rows, and grand totals before upload.
7	Always confirm the correct Month is selected before clicking Upload.
8	Always convert large .xlsx files to .csv for faster processing.
9	Always report unknown GL or material codes from WARNING popups.
10	Always use Google Chrome as the primary browser.

11.2 DON'Ts

Table 10: DON'Ts

#	Do Not
1	Do not upload multi-month SAP exports.
2	Do not edit or rename column headers in SAP export files.
3	Do not upload the same file twice for the same section/category/month/year without checking existing values.
4	Do not delete rows from the export file solely for filtering; system filters handle this.
5	Do not convert SAP PDF reports to Excel via third-party tools.
6	Do not close the browser tab while an upload is in progress.
7	Do not share the Web App URL with unauthorized personnel.
8	Do not change section/year/category/month labels in the Target Sheet without informing the System Administrator.
9	Do not ignore WARNING messages about unknown codes.
10	Do not upload password-protected Excel files.

12 APPENDIX A - REQUIRED SAP COLUMN REFERENCE

Table 11: SAP Column Reference

Column Purpose	Primary Header	Alternate Headers	Used In
Quantity	Quantity	Qty, Billed Quantity	All Sections
Unit of Measure	Unit of Measure	Unit, UOM	All Sections
Movement Type	Movement Type	Movement, MVT	Monthly Production
Plant	Plant	(exact match only)	Monthly Production
Material Code	Material	Material Code, MATNR	Production; Sales (exclusions)

Column Purpose	Primary Header	Alternate Headers	Used In
G/L Account	G/L Account	GL Account, G/L	Monthly Sales, Sale Price
Net Value	Net Value	Net Amount, Currency Value, Company Code Currency Value, Amount, Value	Sale Price

13 APPENDIX B - GL ACCOUNT MASTER LIST

Monthly Sales

Category	Configured GL Accounts
TMT Sales Qty	40000000, 40300300, 40600000
Wire Rod Sales Qty	40010090
Billet Sales Qty	40010000, 40302020
TMT Random	40010070
TMT Chilli	40020030
Binding Wire	40020040

Sale Price

Category	Configured GL Accounts	Excluded Codes	Material
TMT	40000000, 40300300, 40600000	1000000075	
WRD	40010090	None	
Billet	40010000, 40302020	None	
TMT Random	40010070	None	
TMT Chilli	40020030	None	
Binding Wire	40020040	None	

14 APPENDIX C - MATERIAL CODE AND RANGE REFERENCE

Table 14: Material Code Ranges

Category	Type	Value / Range
Billet Production	Single Codes	2000000001, 2000000002
TMT Production (Excl. Chilli & Random)	Range	1000000001 - 1000000040
TMT Production (Excl. Chilli & Random)	Range	1000000121 - 1000000188

Category	Type	Value / Range
TMT Production (Excl. Chilli & Random)	Range	1000000321 - 1000000322
TMT Production (Excl. Chilli & Random)	Range	1000000501 - 1000000510
TMT Production (Excl. Chilli & Random)	Range	1000000615 - 1000000628
Wire Rod Production	Range	1000000323 - 1000000331

15 APPENDIX D - ALLOWED PLANT AND MOVEMENT TYPE CODES

Allowed Plant Codes

Category	Allowed Plant Codes
Billet Production	1001, 1003
TMT Production (Excl. Chilli & Random)	1002, 1004
Wire Rod Production	1004

Allowed Movement Types

Category	Allowed Types	Movement	Notes
Billet Production	101, 102		102 reversals are netted off.
TMT Production (Excl. Chilli & Random)	101, 102		102 reversals are netted off.
Wire Rod Production	101		Reversals not applicable.

16 REVISION HISTORY

Table 17: Revision History

Version	Date	Revised By	Change Description
1.0	February 25, 2026	Amanjyot Singh Bagga	Initial version - full SAP export procedures, file preparation, upload steps, exception handling, and appendices.

This SOP must be reviewed whenever a new GL account, material code/range, plant, or movement type is introduced; when the Target Sheet structure changes; or when a new version of the automation system is deployed.